

# Area in Polar Coordinates



## The polar area formula

- 1 State the formula for the area enclosed by a polar curve  $r = f(\theta)$  between  $\theta = \alpha$  and  $\theta = \beta$ .
  - 2 Find the area enclosed by one full loop of  $r = 3\cos \theta$  (i.e.  $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ ).
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## Area of a full cardioid

- 3 Find the total area enclosed by the cardioid  $r = 1 + \cos \theta$ .
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## Area between two polar curves

- 4 Find the area inside the circle  $r = 3$  but outside the circle  $r = 2$ .
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## Area of one petal of a rose curve

- 5 Find the area of one petal of  $r = 2\sin(2\theta)$ , using the interval  $\theta \in [0, \frac{\pi}{2}]$  where one petal is traced.
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## Setting up area between curves that intersect

- 6 Set up (but do not evaluate) the integral for the area inside both  $r = 2$  and  $r = 2 + 2\cos \theta$ , noting they intersect where  $2 = 2 + 2\cos \theta$ , i.e.  $\theta = \pm\frac{\pi}{2}$ .
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## Area of a full rose curve

- 7 Find the total area enclosed by all 4 petals of  $r = 3\cos(2\theta)$ , using the fact that one petal is traced for  $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ .
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## Area inside a limaçon

- 8 Find the area enclosed by the limaçon  $r = 3 + \cos \theta$  (a curve without an inner loop, since the constant exceeds the coefficient).

- 9 Find the area of the small inner loop of the limaçon  $r = 1 + 2\cos \theta$  (constant less than coefficient, so an inner loop exists), traced for  $\theta \in [\frac{2\pi}{3}, \frac{4\pi}{3}]$ .
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### Setting up area problems (AP-style, no evaluation)

- 10 Set up (but do not evaluate) the integral for the area inside the rose  $r = 2\sin(3\theta)$  that lies in the first quadrant only, for one petal traced on  $\theta \in [0, \frac{\pi}{3}]$ .
- 11 Set up (but do not evaluate) the integral for the area between the circle  $r = 5$  and the cardioid  $r = 5 - 5\sin \theta$  (inside the circle, outside the cardioid), for  $\theta \in [0, \pi]$ .
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### Finding intersection points of polar curves

- 12 Find all points of intersection of  $r = 2\cos \theta$  and  $r = 2\sin \theta$  (both circles of radius 1) by setting the equations equal.
- 13 Explain why checking intersections of two polar curves algebraically (setting  $f(\theta) = g(\theta)$ ) can MISS the pole as an intersection point, and why it must be checked separately.
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### Area using symmetry to simplify computation

- 14 Explain how the symmetry of  $r = 3\cos(2\theta)$  about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.
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### Area review: full worked example

- 15 Find the area enclosed by  $r = \sqrt{\theta}$  for  $\theta \in [0, \pi]$  (a spiral arc closed off by the segment back to the pole).
- 16 Find the area enclosed by  $r = 4\sin \theta$  (a circle) for  $\theta \in [0, \pi]$ , and verify your answer matches the known area formula for a circle of the appropriate radius.
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### Comparing polar area to rectangular area (conceptual)

- 17 Explain why the polar area formula  $A = \frac{1}{2} \int r^2 d\theta$  uses  $r^2$  rather than just  $r$ , by relating it to the area of a thin circular sector.

- 18 A student claims the area between  $r = 1$  and  $r = 2$  (an annulus, full circle) is simply  $\frac{1}{2} \int_0^{2\pi} (2 - 1)^2 d\theta$ . Explain the error and give the correct setup.
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### Multiple choice — easy 15

- 19 Find the area enclosed by  $r = 1$  for  $\theta \in [0, \frac{\pi}{2}]$ .
- a.  $\frac{\pi}{4}$
  - b.  $\frac{\pi}{2}$
  - c.  $-\frac{7\pi}{4}$
  - d.  $\frac{3\pi}{4}$
- 20 Find the area enclosed by  $r = 2$  for  $\theta \in [0, 2\pi]$ .
- a.  $4\pi$
  - b.  $5\pi$
  - c.  $\frac{9\pi}{2}$
  - d.  $8\pi$
- 21 Find the area enclosed by  $r = 3$  for  $\theta \in [0, \frac{3\pi}{2}]$ .
- a.  $\frac{31\pi}{4}$
  - b.  $\frac{27\pi}{2}$
  - c.  $\frac{27\pi}{4}$
  - d.  $\frac{29\pi}{4}$

22 Find the area enclosed by  $r = 7$  for  $\theta \in [0, \pi]$ .

- a.  $\frac{49\pi}{2}$
- b.  $25\pi$
- c.  $\frac{53\pi}{2}$
- d.  $49\pi$

23 Find the area enclosed by  $r = 1$  for  $\theta \in [0, \pi]$ .

- a.  $-\frac{3\pi}{2}$
- b.  $\pi$
- c.  $\frac{5\pi}{2}$
- d.  $\frac{\pi}{2}$

24 Find the area enclosed by  $r = 6$  for  $\theta \in [0, 2\pi]$ .

- a.  $\frac{73\pi}{2}$
- b.  $36\pi$
- c.  $37\pi$
- d.  $72\pi$

25 Find the area enclosed by  $r = 6$  for  $\theta \in [0, \frac{\pi}{2}]$ .

- a.  $18\pi$
- b.  $7\pi$
- c.  $\frac{19\pi}{2}$
- d.  $9\pi$

26 Find the area enclosed by  $r = 6$  for  $\theta \in [0, \frac{3\pi}{2}]$ .

a.  $27\pi$

b.  $\frac{55\pi}{2}$

c.  $26\pi$

d.  $54\pi$

27 Find the area enclosed by  $r = 9$  for  $\theta \in [0, \frac{\pi}{2}]$ .

a.  $\frac{81\pi}{4}$

b.  $\frac{81\pi}{2}$

c.  $\frac{89\pi}{4}$

d.  $\frac{83\pi}{4}$

28 Find the area enclosed by  $r = 3$  for  $\theta \in [0, 2\pi]$ .

a.  $9\pi$

b.  $18\pi$

c.  $\frac{19\pi}{2}$

d.  $7\pi$

29 Find the area enclosed by  $r = 1$  for  $\theta \in [0, \frac{3\pi}{2}]$ .

a.  $\frac{5\pi}{4}$

b.  $\frac{3\pi}{4}$

c.  $\frac{7\pi}{4}$

d.  $\frac{3\pi}{2}$

30 Find the area enclosed by  $r = 9$  for  $\theta \in [0, \pi]$ .

a.  $81\pi$

b.  $\frac{83\pi}{2}$

c.  $\frac{81\pi}{2}$

d.  $41\pi$

31 Find the area enclosed by  $r = 8$  for  $\theta \in [0, \frac{\pi}{2}]$ .

a.  $\frac{33\pi}{2}$

b.  $17\pi$

c.  $16\pi$

d.  $32\pi$

32 Find the area enclosed by  $r = 8$  for  $\theta \in [0, 2\pi]$ .

a.  $\frac{129\pi}{2}$

b.  $62\pi$

c.  $64\pi$

d.  $128\pi$

33 Find the area enclosed by  $r = 4$  for  $\theta \in [0, \frac{3\pi}{2}]$ .

a.  $11\pi$

b.  $\frac{25\pi}{2}$

c.  $24\pi$

d.  $12\pi$

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### Multiple choice — medium 15

- 34 Find the area inside the circle  $r = 9$  but outside the circle  $r = 4$ .
- a.  $-7\pi$
  - b.  $25\pi$
  - c.  $77\pi$
  - d.  $65\pi$
- 35 Find the area inside the circle  $r = 13$  but outside the circle  $r = 5$ .
- a.  $-12\pi$
  - b.  $64\pi$
  - c.  $144\pi$
  - d.  $164\pi$
- 36 Find the area inside the circle  $r = 8$  but outside the circle  $r = 5$ .
- a.  $39\pi$
  - b.  $59\pi$
  - c.  $9\pi$
  - d.  $-17\pi$
- 37 Find the area inside the circle  $r = 10$  but outside the circle  $r = 6$ .
- a.  $-26\pi$
  - b.  $64\pi$
  - c.  $94\pi$
  - d.  $16\pi$

- 38 Find the area inside the circle  $r = 4$  but outside the circle  $r = 2$ .
- a.  $14\pi$
  - b.  $12\pi$
  - c.  $4\pi$
  - d.  $0\pi$
- 39 Find the area inside the circle  $r = 13$  but outside the circle  $r = 6$ .
- a.  $163\pi$
  - b.  $-23\pi$
  - c.  $49\pi$
  - d.  $133\pi$
- 40 Find the area inside the circle  $r = 13$  but outside the circle  $r = 7$ .
- a.  $162\pi$
  - b.  $-36\pi$
  - c.  $120\pi$
  - d.  $36\pi$
- 41 Find the area inside the circle  $r = 10$  but outside the circle  $r = 5$ .
- a.  $-15\pi$
  - b.  $75\pi$
  - c.  $95\pi$
  - d.  $25\pi$

- 42 Find the area inside the circle  $r = 15$  but outside the circle  $r = 7$ .
- a.  $64\pi$
  - b.  $176\pi$
  - c.  $-34\pi$
  - d.  $218\pi$
- 43 Find the area inside the circle  $r = 8$  but outside the circle  $r = 3$ .
- a.  $55\pi$
  - b.  $61\pi$
  - c.  $25\pi$
  - d.  $-\pi$
- 44 Find the area inside the circle  $r = 3$  but outside the circle  $r = 2$ .
- a.  $\pi$
  - b.  $-\pi$
  - c.  $5\pi$
  - d.  $7\pi$
- 45 Find the area inside the circle  $r = 8$  but outside the circle  $r = 1$ .
- a.  $49\pi$
  - b.  $7\pi$
  - c.  $61\pi$
  - d.  $63\pi$

46 Find the area inside the circle  $r = 10$  but outside the circle  $r = 3$ .

- a.  $49\pi$
- b.  $97\pi$
- c.  $91\pi$
- d.  $\pi$

47 Find the area inside the circle  $r = 5$  but outside the circle  $r = 4$ .

- a.  $9\pi$
- b.  $\pi$
- c.  $-11\pi$
- d.  $21\pi$

48 Find the area inside the circle  $r = 11$  but outside the circle  $r = 8$ .

- a.  $9\pi$
- b.  $-53\pi$
- c.  $113\pi$
- d.  $57\pi$

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### Multiple choice — hard 15

49 Find the total area enclosed by the curve  $r = 8 + 7\sin \theta$ .

- a.  $113\pi$
- b.  $\frac{177\pi}{2}$
- c.  $\frac{179\pi}{2}$
- d.  $\frac{225\pi}{2}$

50 Find the total area enclosed by the curve  $r = 7 + 6\cos \theta$ .

- a.  $85\pi$
- b.  $68\pi$
- c.  $\frac{169\pi}{2}$
- d.  $67\pi$

51 Find the total area enclosed by the curve  $r = 5 + 9\sin \theta$ .

- a.  $98\pi$
- b.  $106\pi$
- c.  $\frac{133\pi}{2}$
- d.  $\frac{131\pi}{2}$

52 Find the total area enclosed by the curve  $r = 4 + 9\cos \theta$ .

- a.  $\frac{169\pi}{2}$
- b.  $97\pi$
- c.  $\frac{115\pi}{2}$
- d.  $\frac{113\pi}{2}$

53 Find the total area enclosed by the curve  $r = 3 + 5\cos \theta$ .

- a.  $34\pi$
- b.  $32\pi$
- c.  $\frac{45\pi}{2}$
- d.  $\frac{43\pi}{2}$

54 Find the area of one petal of  $r = 4\cos(2\theta)$ .

- a.  $\frac{9\pi}{4}$
- b.  $2\pi$
- c.  $4\pi$
- d.  $8\pi$

55 Find the area of one petal of  $r = 4\sin(7\theta)$ .

- a.  $\frac{23\pi}{28}$
- b.  $\frac{8\pi}{7}$
- c.  $\frac{16\pi}{7}$
- d.  $\frac{4\pi}{7}$

56 Find the area of one petal of  $r = 2\sin(5\theta)$ .

- a.  $\frac{4\pi}{5}$
- b.  $\frac{7\pi}{10}$
- c.  $\frac{2\pi}{5}$
- d.  $\frac{\pi}{5}$

57 Find the area of one petal of  $r = 3\cos(6\theta)$ .

- a.  $\frac{3\pi}{2}$
- b.  $\frac{3\pi}{8}$
- c.  $\frac{3\pi}{4}$
- d.  $\frac{7\pi}{8}$

58 Find the area of one petal of  $r = 4\sin(2\theta)$ .

a.  $\frac{5\pi}{2}$

b.  $8\pi$

c.  $4\pi$

d.  $2\pi$

59 Find the area inside the circle  $r = 9$  but outside the circle  $r = 2$ , for  $\theta \in [0, \pi]$  only (half the full region).

a.  $77\pi$

b.  $\frac{79\pi}{2}$

c.  $\frac{77\pi}{2}$

d.  $\frac{49\pi}{2}$

60 Find the area inside the circle  $r = 11$  but outside the circle  $r = 7$ , for  $\theta \in [0, \pi]$  only (half the full region).

a.  $8\pi$

b.  $72\pi$

c.  $36\pi$

d.  $37\pi$

61 Find the area inside the circle  $r = 3$  but outside the circle  $r = 2$ , for  $\theta \in [0, \pi]$  only (half the full region).

a.  $\frac{7\pi}{2}$

b.  $\frac{5\pi}{2}$

c.  $\frac{\pi}{2}$

d.  $5\pi$

- 62 Find the area inside the circle  $r = 3$  but outside the circle  $r = 1$ , for  $\theta \in [0, \pi]$  only (half the full region).
- a.  $5\pi$
  - b.  $8\pi$
  - c.  $4\pi$
  - d.  $2\pi$
- 63 Find the area inside the circle  $r = 13$  but outside the circle  $r = 5$ , for  $\theta \in [0, \pi]$  only (half the full region).
- a.  $72\pi$
  - b.  $73\pi$
  - c.  $32\pi$
  - d.  $144\pi$